

Functional Annotation Report: *cyoB* (PP_0813, UniProt Q88PN6)

Cytochrome *bo*₃ ubiquinol oxidase subunit I — *Pseudomonas putida* KT2440

1. Summary (Answer to the Research Question)

cyoB (locus PP_0813, UniProt Q88PN6) encodes subunit I of the cytochrome *bo*₃ ubiquinol oxidase (EC 7.1.1.3), a member of the heme-copper respiratory oxidase superfamily. This subunit is the catalytic core of a terminal oxidase of the aerobic respiratory chain. It binds all of the enzyme's redox-active metal centers — a low-spin heme *b* and the high-spin heme *o*₃-Cu_B binuclear center — and there it carries out the enzyme's primary chemistry: oxidation of ubiquinol (2 e⁻) coupled to the four-electron reduction of molecular O₂ to water, using the released free energy to pump protons across the cytoplasmic (inner) membrane and generate proton-motive force. In *P. putida* KT2440 the enzyme (Cyo) is the dominant terminal oxidase during aerobic exponential growth and additionally acts as a respiratory redox sensor that feeds into global metabolic gene regulation.

The target identity is unambiguous and fully consistent with the UniProt record: gene symbol, EC number, protein family, and InterPro domains all match the primary literature on cytochrome *bo*₃/*cyo*.

2. Target Identity Verification

Attribute	UniProt record	Verified in this study
Gene symbol	<i>cyoB</i>	✓ Matches subunit I of <i>cyo</i> operon
Locus	PP_0813	✓ Confirmed within PP_0812–PP_0816 <i>cyoABCDE</i> cluster
Product	Cytochrome <i>bo</i> (3) ubiquinol oxidase subunit 1	✓ Consistent with all literature
EC	7.1.1.3	✓ Ubiquinol:O ₂ oxidoreductase (proton-pumping)
Family	Heme–copper respiratory oxidase	✓ A-type oxygen reductase
Organism	<i>P. putida</i> KT2440 (taxid 160488)	✓

Bioinformatic confirmation. A Needleman–Wunsch alignment of Q88PN6 (672 aa) against *E. coli* CyoB (P0ABI8) gave **69.1 % amino-acid identity**, and **all six catalytic metal-ligand histidines were strictly conserved at identical positions**: His106/His421 (low-spin heme *b*), His419 (high-spin heme *o*₃), and His284/His333/His334 (Cu_B). The **D-channel proton-pumping glutamate Glu286** was likewise strictly conserved, as were the **active-site cross-linked tyrosine Tyr288** (covalently linked to the Cu_B ligand His284; supplies the fourth electron/proton for O–O bond scission) and the **K-pathway lysine Lys362** — i.e., the entire canonical A-type catalytic and proton-input apparatus is intact in the *P. putida* protein. UniProt predicts **14 transmembrane helices** and a "Cytochrome oxidase subunit I profile" domain (res 39–559), matching the polytopic subunit-I fold. This conservation licenses transfer of the detailed, experimentally derived *E. coli* mechanism to the *P. putida* protein.

The gene symbol *cyoB* is **not** ambiguous here; abundant primary literature on cytochrome *bo*₃ (chiefly from *E. coli*, plus *P. putida*-specific studies) directly describes this exact protein.

3. Primary Function: The Catalyzed Reaction and Substrate Specificity

3.1 Reaction

Cytochrome *bo*₃ catalyzes the **two-electron oxidation of ubiquinol and the reduction of molecular oxygen to water** [Welter et al. 1994,

P 7961841]



i.e., four electrons drawn from two ubiquinol molecules fully reduce O₂ to two H₂O, while additional protons are vectorially translocated (see §4). Subunit I (CyoB) is where the O₂ chemistry occurs: **"subunit I contains the binuclear center where oxygen is reduced to water"** [Park et al. 2002,

P 12080058]

The center comprises **heme B at the low-spin site and heme O at the high-spin site of the binuclear center** [Uchida et al. 2000,

P 11132640]

3.2 Substrate specificity

The physiological electron-donating substrate is **ubiquinol** — the reduced form of the membrane isoprenoid quinone pool (ubiquinol-8 in *E. coli*; *Pseudomonas* species use ubiquinone-9 as their major quinone). The enzyme also harbors a **single, tightly bound high-affinity ubiquinone-8** that is a genuine redox cofactor: because cytochrome *bo*₃ **lacks the Cu_A center** of cytochrome *c* oxidases, this bound quinone acts as the **"fourth electron donor (a Cu_A substitute)"** that rapidly re-reduces low-spin heme *b* and lets the O₂ reaction run to completion instead of stalling at the ferryl intermediate; reconstitution with ubiquinone-8 converts monophasic to multiphasic kinetics [Puustinen et al. 1996,

P 8643669]

Thus there are effectively two quinone-binding sites (high- and low-affinity) [Musser & Chan 1998,

],

P 9545462

and photoaffinity labelling with azido-ubiquinone localizes quinol binding to the subunit I/II region [Welter et al. 1994,

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P 7961841

Abramson et al. 2000,

].

P 11017202

3.3 Structural basis of the active site

The first structure of a ubiquinol oxidase — *E. coli* cytochrome *bo*₃ — revealed that **the membrane-spanning region of subunit I contains a cluster of polar residues exposed to the lipid bilayer that forms a quinone-binding site**, and mutagenesis supports specific roles for these residues in electron/proton transfer [Abramson et al. 2000,

].

P 11017202

The metal centers all reside in subunit I, the **common functional core** shared with cytochrome *c* oxidases [Gohlke et al. 1997,

].

P 9135135

4. Mechanism: Proton Pumping and Energy Conservation

Cytochrome *bo*₃ is an **A-type heme–copper oxygen reductase**. Proton transfer to the catalytic site uses **two well-defined pathways, the D- and K-pathways** [Lee et al. 2012,

].

P 22056517

The **D-pathway terminates at the strictly conserved Glu286** (helix VI of subunit I), which controls proton delivery **both to the active site for O₂ chemistry and to a proton-loading site for pumping**; mutation of E286 blocks proton flux and abolishes turnover, exclusively by inhibiting proton transfer in the *bo*₃ enzyme [Egawa et al. 2011,

];

P 21684251

Svensson-Ek et al. 1996,

P 8885847

Time-resolved FTIR shows redox-linked perturbation of E286 during the catalytic cycle, implicating it directly in proton pumping [Bailey et al. 2002,

P 11851414

The net result is an **electrogenic, redox-driven proton pump**: electrons flow from ubiquinol through heme *b* to the heme *o*₃-Cu_B center to reduce O₂, while protons are pumped from the cytoplasm to the periplasm, contributing to the proton-motive force that powers ATP synthesis, transport, and motility. A quinol (Q-loop) contribution from the two quinone sites adds further charge separation unique to quinol oxidases [Musser & Chan 1998,

P 9545462

5. Holoenzyme Architecture and Cofactor Biosynthesis

CyoB is subunit I of a **four-subunit complex**. Partial sequencing of the purified *E. coli* enzyme identified subunits I, II and III as the **cyoB, cyoA and cyoC gene products** (~58, 33, 22 kDa), plus subunit IV (cyoD, ~17 kDa) [Minghetti et al. 1992,

P 1322173

Metal analysis shows **2 equivalents of iron (heme *b* + heme *o*₃) but only 1 equivalent of copper (Cu_B); the Cu_A center is absent** [Minghetti et al. 1992,

P 1322173

Accessory-subunit studies show **subunit IV (CyoD) is required for Cu_B insertion into subunit I**, acting as a domain-specific assembly chaperone that contacts helices VII/VIII (the Cu_B-binding region) of subunit I [Saiki et al. 1997,

P 9169436

Operon and cofactor supply (bioinformatic, *P. putida*-specific). A UniProt query of *P. putida* KT2440 resolves a **syntenic cyoABCDE operon**: PP_0812 *cyoA* (subunit II, Q88PN7) – **PP_0813 cyoB (subunit I, Q88PN6)** – PP_0814 *cyoC* (subunit III, Q88PN5) – PP_0815 *cyoD* (subunit IV, Q88PN4) – PP_0816 *cyoE2* (**protoheme IX farnesyltransferase / heme-O synthase**, Q88PN3). CyoE converts protoheme IX to **heme O**, the cofactor inserted into subunit I's high-spin (*o*₃) site

[Minghetti et al. 1992,]
 P 1322173

The operon organization matches the *E. coli* prototype, corroborating that CyoB assembles and functions identically.

6. Localization

CyoB carries out its function **embedded in the cytoplasmic (inner) membrane**. UniProt predicts **14 helical transmembrane spans** (res 16–604), consistent with a deeply membrane-embedded catalytic subunit. The cytochrome *bo*₃ complex is an inner-membrane protein assembly: its subunit II (CyoA) is integrated/translocated across the *E. coli* inner membrane via the **YidC and SecYEG** pathways [van Bloois et al. 2006,]
 P 16481320

Celebi et al. 2008,]
 P 18155041

The hemes and Cu_B of subunit I lie within the membrane; ubiquinol (a lipid-soluble substrate) is oxidized at a membrane-buried Q-site, and protons are translocated vectorially across the same membrane (cytoplasm → periplasm).

7. Biological Process / Pathway Context in *P. putida* KT2440

P. putida KT2440 has a **branched aerobic respiratory chain with multiple terminal oxidases** — Cyo (*bo*₃), CIO (cyanide-insensitive), *cbb*₃-1, *cbb*₃-2 and *aa*₃ — whose relative use varies with O₂ and growth phase [Morales et al. 2006,]
 P 16958757

Ugidos et al. 2008,]
 P 18341582

Functional assays show **Cyo has a leading role during aerobic exponential growth**, while *cbb*₃-1 dominates in stationary phase/low O₂ [Ugidos et al. 2008,

].

P 18341582

Expression is oxygen-responsive and coordinated by the **ANR global regulator**, which binds the *cyo* promoter [Ugidos et al. 2008,

].

P 18341582

Beyond bioenergetics, the Cytochrome oxidase acts as a **respiratory signal transmitter**: inactivation of *cyo* has little effect on growth rate but **broadly remodels the transcriptome and relieves the global (catabolite) control that represses alkane-degradation genes**, identifying *cyo* specifically as "**the oxidase that transmits a regulatory signal related to the activity of the electron transport chain**"; loss of *cyo* is compensated by up-regulation of the CIO and *cbb₃-1* oxidases [Morales et al. 2006,

].

P 16958757

This links Cytochrome B's catalytic (electron-transport) activity to metabolic gene regulation in this environmentally and biotechnologically important soil bacterium.

8. Evidence Summary

Claim	Evidence type	Source
Subunit I holds the O ₂ -reducing binuclear center	Biochemistry / two-hybrid	P 12080058
Cofactors: low-spin heme <i>b</i> + high-spin heme <i>o</i> ₃	EPR / Raman	P 11132640
Reaction: 2e ⁻ ubiquinol oxidation + O ₂ →H ₂ O	Enzymology	P 7961841
Quinone-binding site in subunit I	X-ray structure + mutagenesis	P 11017202
Bound Q8 substitutes for Cu_A	Flow-flash kinetics	P 8643669
Glu286 required for proton transfer/pumping	Mutagenesis / FTIR	P 8885847
		P 21684251
		P 11851414
A-type oxidase; D-/K-pathways	Review	P 22056517
Six conserved His in subunit I	Comparative / structural	P 9545462 ; this study (69.1 % id, all 6 His conserved)
Active-site Tyr288 + K-pathway Lys362 conserved	Sequence alignment	This study
2 Fe + 1 Cu, no Cu_A; four subunits = cyoBAC(D)	Metal analysis / sequencing	P 1322173
CyoD chaperones Cu_B insertion	Mutagenesis	P 9169436
Inner-membrane localization	Biogenesis studies	P 16481320
		P 18155041

Claim	Evidence type	Source
Dominant oxidase in aerobic exponential growth (<i>P. putida</i>)	Functional assays	P 18341582
Respiratory signalling role (<i>P. putida</i>)	Microarray / RT-PCR	P 16958757
cyoABCDE operon = PP_0812–0816	Genomic query	This study (UniProt)

9. Supported and Refuted Hypotheses

Supported - CyoB is the catalytic subunit I of cytochrome *bo*₃ ubiquinol oxidase (identity confirmed by annotation, 69.1 % identity to *E. coli* CyoB, and strict conservation of all six metal ligands + Glu286). - Primary function = ubiquinol:O₂ oxidoreductase that reduces O₂ to water and pumps protons. - Localizes to and functions within the inner membrane. - In *P. putida*, it is the principal aerobic-growth terminal oxidase and a respiratory regulatory sensor.

Refuted / excluded - CyoB is **not** a cytochrome *c* oxidase (no Cu_A; electrons enter from ubiquinol, not cytochrome *c*). - The gene is **not** ambiguous or mis-assigned; no competing gene with the same symbol confounds the annotation.

10. Limitations and Future Directions

- Most mechanistic/structural data derive from the orthologous *E. coli* enzyme; no *P. putida* CyoB crystal/cryo-EM structure or direct enzymology was found. The very high conservation makes transfer of mechanism robust, but organism-specific kinetic parameters (e.g., quinol specificity for ubiquinol-9, O₂ affinity) remain to be measured directly.
- The precise molecular link between Cyo electron-transport activity and the alkane-degradation regulatory signal in *P. putida* is phenomenological; the downstream signalling components are not fully defined.

- Proton-pumping stoichiometry (H^+/e^-) was not directly quantified here; the A-type consensus (~1 pumped H^+ per electron, plus scalar/chemical protons) is inferred from the superfamily.

Report compiled from primary literature and bioinformatic analysis (sequence alignment and genome operon mapping). All quantitative sequence claims were computed in this study against UniProt reference sequences Q88PN6 and P0ABI8.

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